2025 BATCH

PRINCIPLES OF DATA SCIENCE

CW 2 (PORTFOLIO) & GROUP WORK-7144

COVENTRY UNIVERSITY
MSC DATA SCIENCE

Portfolio CW 2 :

Question 1

Instructions:
- Select ONE of the following two questions.
- Your project must include Data Collection, EDA, Problem Solving, Cloud Integration (AWS/Azure), Git Collaboration, and Final Reporting.
- All work must be managed in a Git repository with meaningful commits.

OPTION 1: Agentic AI Workflow for Automated Trading (Finance Domain)

Problem Statement:
Design and analyze an Agentic AI-based trading workflow system that can autonomously perform market analysis, retrieve financial information, and execute buy/sell decisions.
Your project must demonstrate how data-driven insights can support automated trading strategies.

1. Data Collection
- Collect financial market data (stocks/crypto/options) using APIs (Yahoo Finance, Alpha Vantage, Binance API), web scraping, or public datasets.
- Clearly explain your data collection method.
- Integrate Cloud Services (AWS/Azure):
  - AWS S3 / Azure Blob Storage for storing raw data
  - AWS EC2 / Azure VM for computation
  - AWS RDS / Azure SQL Database for structured storage
  - Optional: AWS Lambda / Azure Functions for automation

2. Exploratory Data Analysis (EDA)
- Data cleaning and preprocessing
- Handling missing values and anomalies
- Descriptive statistics and time-series visualization
- Volatility and correlation analysis
- Clustering of assets (e.g., K-Means)
- Feature engineering (Moving Averages, RSI, MACD)
- Highlight market patterns, risk indicators, and Buy/Sell signals

3. Problem Solving
- Identify a trading-related problem.
- Propose an Agentic AI Workflow Architecture including:
  - Market Analysis Module
  - Information Retrieval Module (internet/news/database)
  - Decision Engine (Buy/Sell/Hold)
  - Risk Management Module
- Evaluate model accuracy and backtesting performance.
- Discuss business/financial impact.

4. Collaboration & Documentation
- Use Git for version control with meaningful commits.
- Document architecture diagrams, cloud setup, model results, insights, and limitations.

OPTION 2: Real-World Data Analysis & Business Problem Solving (Open Domain)

Problem Statement:
Select a real-world dataset and perform a complete data analysis lifecycle to identify patterns, build predictive models, and propose a solution to a business problem.
Examples include customer churn prediction, sales forecasting, fraud detection, healthcare analytics, e-commerce recommendation systems, or transportation demand analysis.

1. Data Collection
- Use public datasets (Kaggle, UCI, Government portals), APIs, web scraping, or surveys.
- Clearly explain the collection methodology.
- Integrate Cloud Services (AWS/Azure):
  - AWS S3 / Azure Blob Storage
  - AWS EC2 / Azure ML / Azure VM for computation
  - AWS RDS / Azure SQL for storage
  - Optional: AWS SageMaker or Azure ML for deployment

2. Exploratory Data Analysis (EDA)
- Data cleaning & preprocessing
- Handling missing values and outliers
- Descriptive statistics and visualizations
- Correlation heatmaps
- Feature engineering
- Clustering analysis (e.g., customer segmentation)
- Identification of trends and business insights

3. Problem Solving
- Clearly define a business problem.
- Build a predictive or analytical model.
- Evaluate performance (Accuracy, Precision/Recall, RMSE if regression).
- Discuss business impact, scalability, and deployment considerations.

4. Collaboration & Documentation
- Use Git for project management.
- Maintain structured commits.
- Document decisions, code, assumptions, challenges, and cloud setup.

Final Deliverables (For Both Options)

1. Well-structured Git Repository
2. ETL and Analysis Notebooks/Scripts
3. Final Report including:
   - Dataset insights

- Clustering analysis results
- Model performance metrics
- Business impact analysis
- Operational improvement recommendations
- Cloud architecture overview (AWS/Azure)

Question 2:

Compare and contrast the following probability distributions, in detail.

a. Binomial distribution
b. Possin distribution
c. Geometric distribution
d. Hypergeometric distribution
e. Negative binomial distribution
f. Exponential distribution
g. Normal distribution
h. Chi square distribution

Group Assignment

Instructions:
- Use the dataset provided.
- The objective is to analyze and improve operations using data-driven insights.
- Your project must include ETL, Clustering, Forecasting, Git Collaboration, and Final Presentation.
- All work must be version-controlled using Git with meaningful commits.

Problem Statement

You are tasked with analyzing and improving operations for the given dataset. Need better understand its operational performance, customer satisfaction, and demand trends to optimize efficiency and strategic decision-making.

Project Requirements

1. ETL Process
- Perform a complete ETL (Extract, Transform, Load) process.
- Clean missing values, remove inconsistencies, and handle outliers.
- Transform and integrate datasets for analysis.
- Document the entire ETL pipeline clearly (tools, steps, assumptions).

2. Exploratory Data Analysis (EDA)
- Perform detailed exploratory data analysis.
- Generate descriptive statistics.
- Create visualizations to identify trends, correlations, and anomalies.
- Highlight key operational and customer insights.

3. Clustering Analysis
- Apply clustering techniques (e.g., K-Means, Hierarchical Clustering).
- Identify meaningful groupings such as:
  - Customer Segments
  - Service Performance Categories
  - Demand Patterns
- Interpret clustering results and discuss business implications.

4. Forecasting Model
- Develop a forecasting model to predict future trends.
- Examples: Sales performance, customer satisfaction scores, order volumes.
- Evaluate model performance using appropriate metrics (MAE, RMSE, Accuracy, etc.).
- Discuss potential operational impact of forecasts.

5. Git Repository & Documentation
- Use Git for version control.
- Maintain structured commits with meaningful messages.
- Organize repository with clear folder structure.
- Document decisions, assumptions, and challenges.

Work Division
Divide the following tasks equally among team members:

1. Data Preprocessing (ETL)
2. Exploratory Data Analysis (EDA)
3. Problem Identification and Solution Proposal
4. Git Repository Management and Documentation

Ensure equal participation and contribution from all members.

Submission Requirements
Submit the following:

1. Git repository link.
2. Analysis notebooks/scripts demonstrating ETL, clustering, and forecasting.
3. A brief presentation summarizing:
   - Key insights and trends
   - Clustering results and business interpretation
   - Forecasting model performance

- Operational recommendations
- Limitations and future improvements